



IOT-ENABLED PREDECTIVE MAINTENANCE OF SOLAR PANELS USING MACHINE LEARNING FOR ANOMALY DETECTION AND PERFORMANCE FORECASTING.

OUR TEAM:

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PROBLEM + STATEMENT

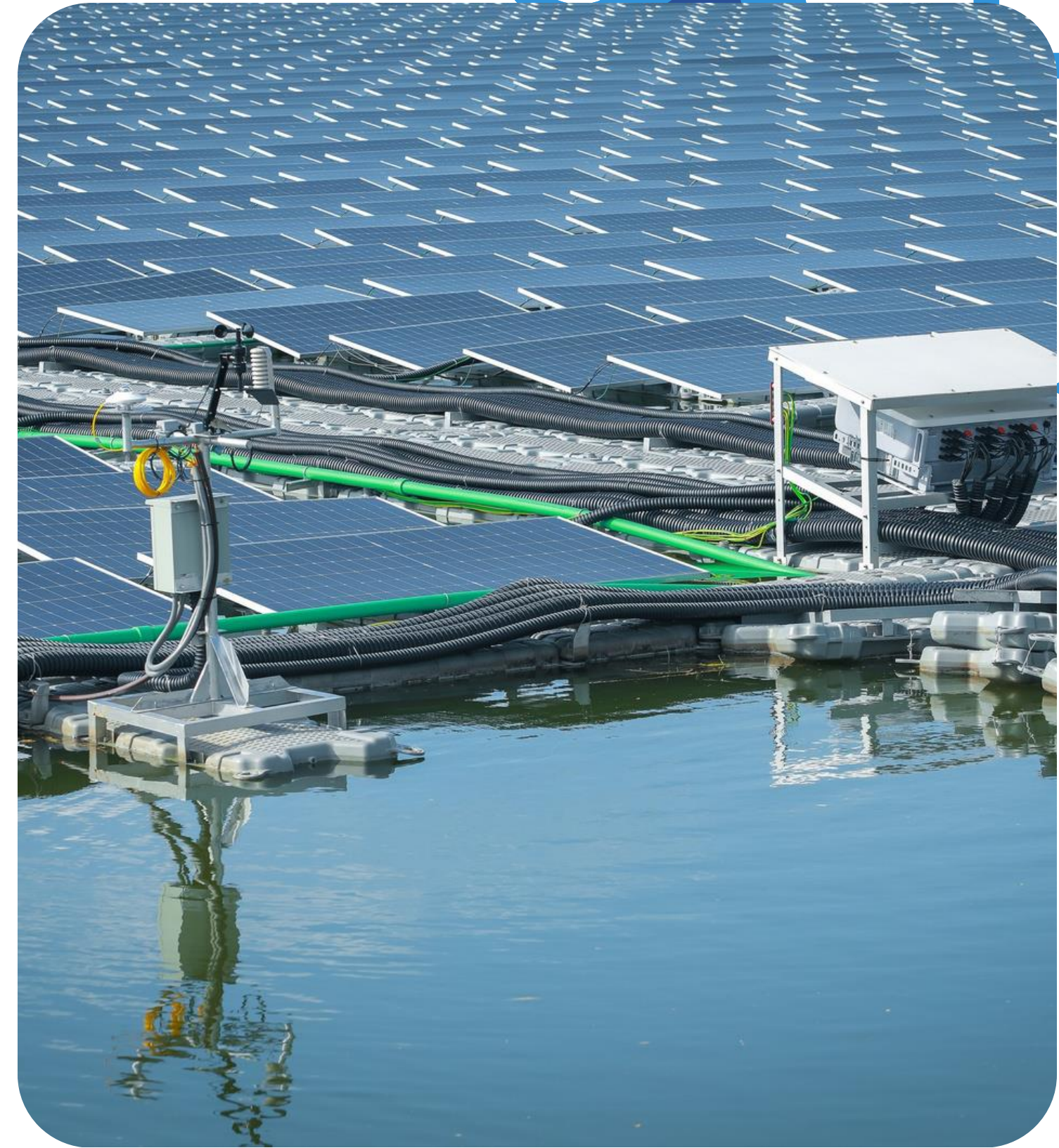
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Solar panel performance decreases due to dust, shading, high temperature, wiring faults, etc. Existing maintenance is either: Reactive (fix only after failure), Time-based (routine checks even if no issue). This results in: Energy loss Increased maintenance cost No early fault detection



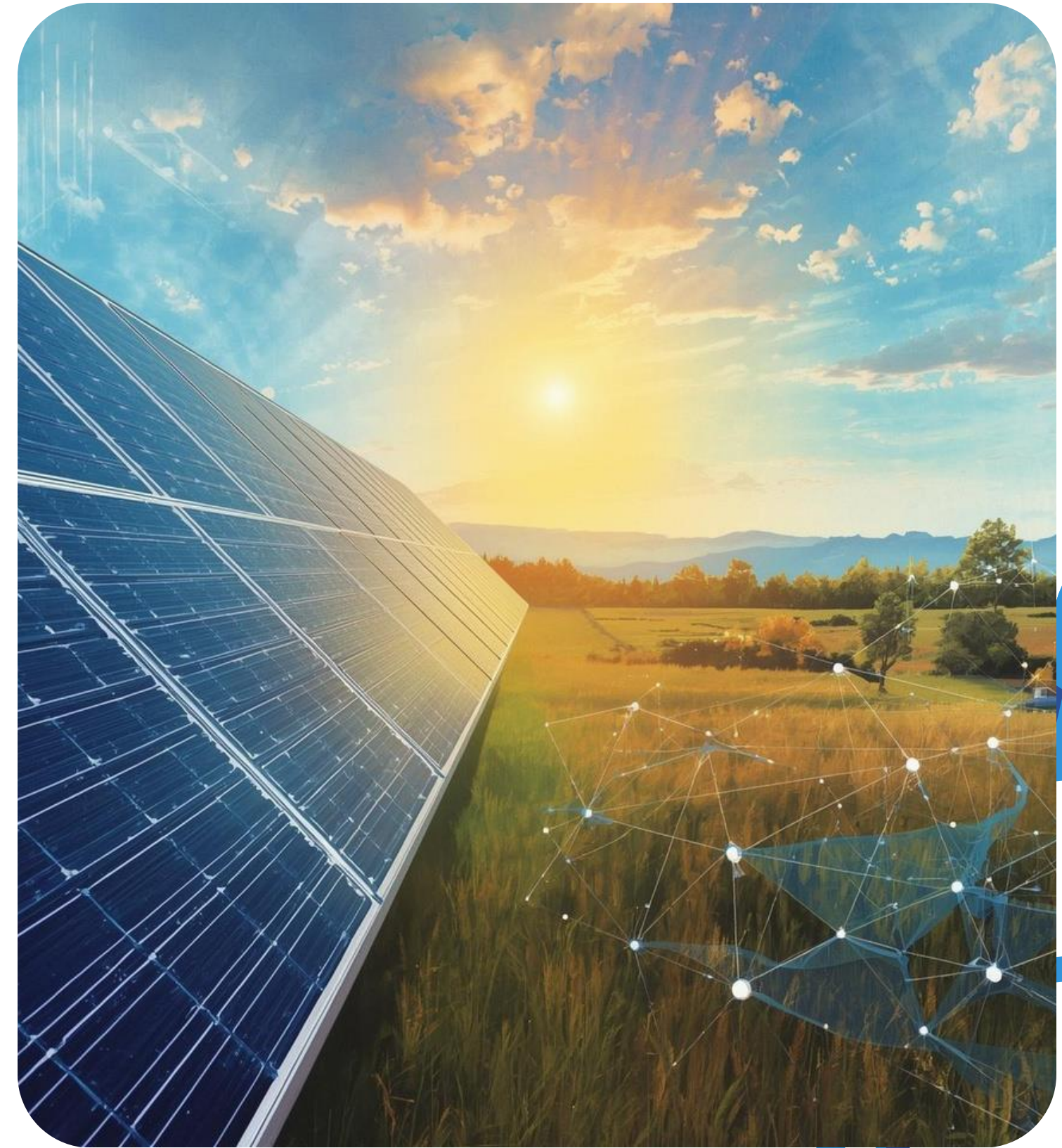
AIMS & OBJECTIVES

Aim: Develop a system that predicts power output and detects faults using IoT + ML. **Objectives:** Collect data through IoT sensors (irradiance, voltage, current, temperature). Predict performance using ML algorithms. Detect anomalies before failures occur. Reduce manual inspection and improve maintenance efficiency.



BACKGROUND & BASIC CONCEPT

Solar panel efficiency varies with sunlight, temperature, dust accumulation. IoT enables real-time monitoring of the solar system. ML learns patterns from data and predicts: Expected power output Abnormal values → fault alert



DATASET OVERVIEW

Dataset contains real-time solar parameters:

Irradiance (W/m^2)

Voltage (V)

Current (A)

Temperature ($^{\circ}\text{C}$)

Power output (W)

Dataset is preprocessed:

Handling missing values

Normalization Train/Test
split (80:20)

ALGORITHMS USED

1. Random Forest

Best for multi-class fault detection

Handles non-linear and noisy sensor data

High accuracy and stable results

2. Gradient Boosting

Learns from previous errors

Captures small variations in voltage/current

Improves prediction quality

3. Isolation Forest

Detects anomalies without labelled data

Finds sudden power/voltage deviations

Ideal for real-time IoT monitoring

METHODOLOGY



IoT sensors collect data from solar panels



Data stored in cloud



Preprocessing & feature extraction



Train ML models



Predict output + detect anomalies



Display results on dashboard

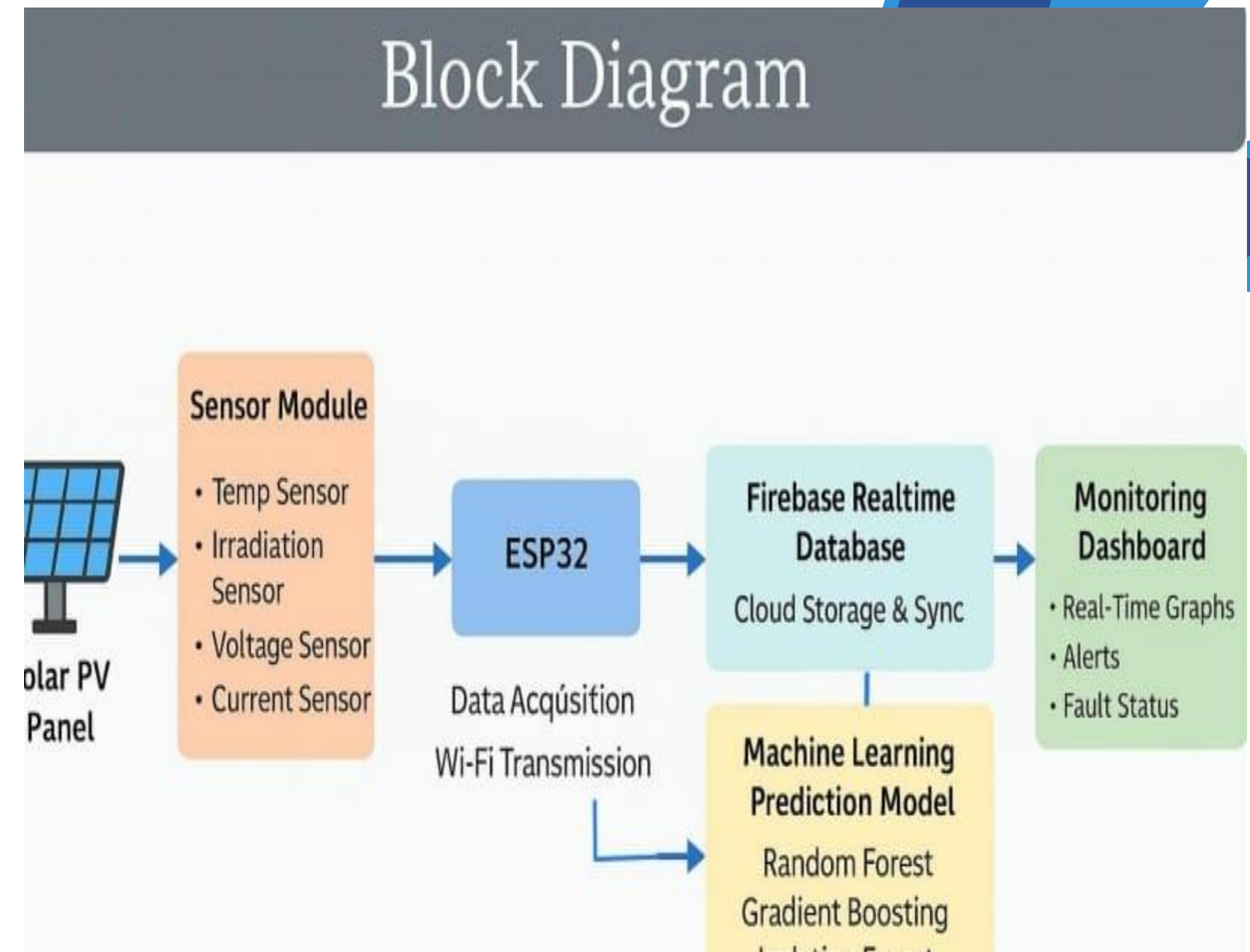
BLOCK DIAGRAM

The sensors (temperature, irradiance, voltage, current) collect data from the solar panel.

The ESP32 receives this data and sends it to the Firebase cloud through Wi-Fi.

The ML model in the cloud checks the data and identifies if the panel is normal or faulty.

The dashboard shows live readings and fault alerts to the user.



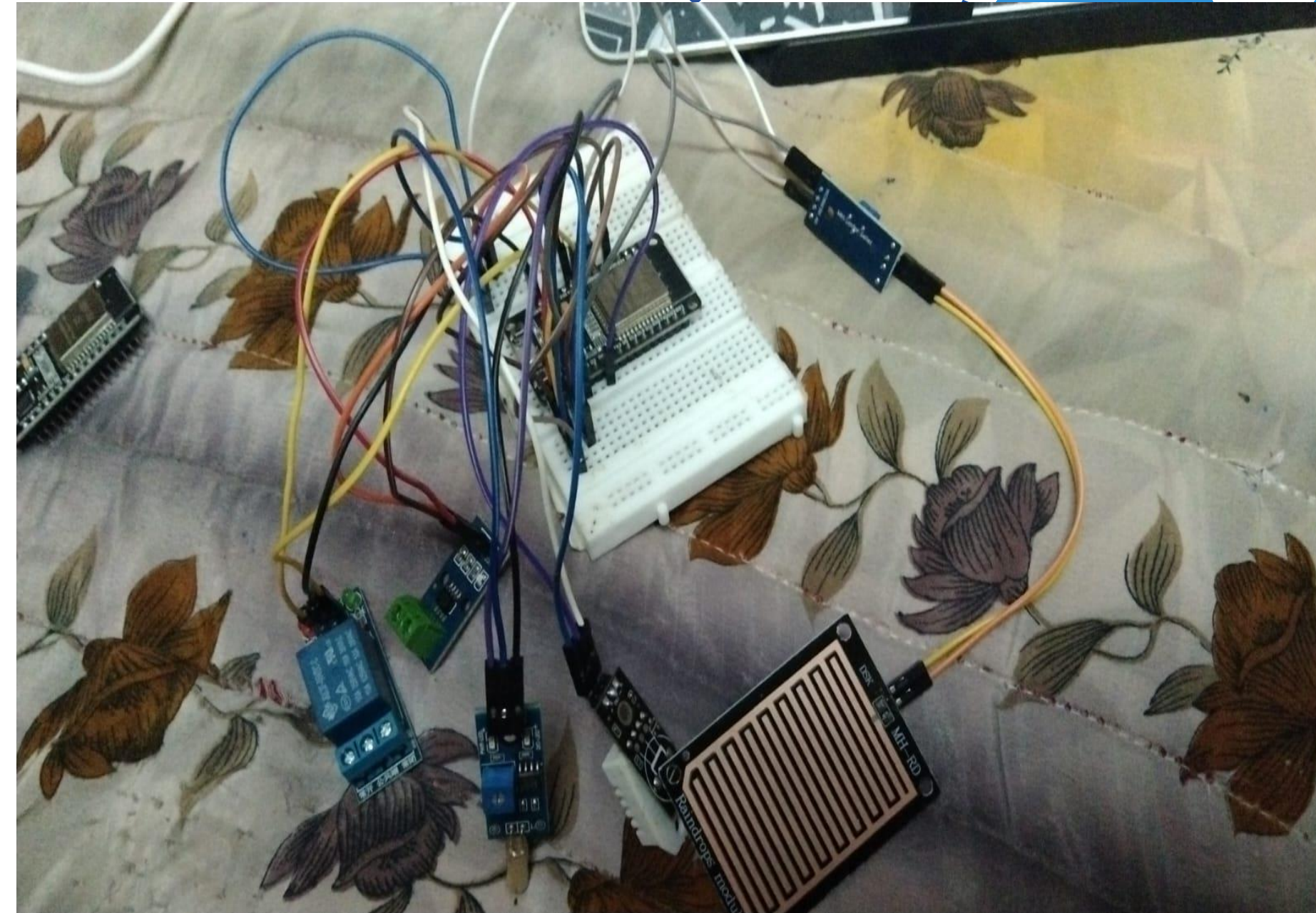
Hardware Connection

The solar panel is connected to the voltage sensor and current sensor so we can measure power.

All sensors (temperature, LDR, voltage, current) are connected to the ESP32 using their signal pins. Every sensor's GND is connected to the ESP32 GND to complete the circuit.

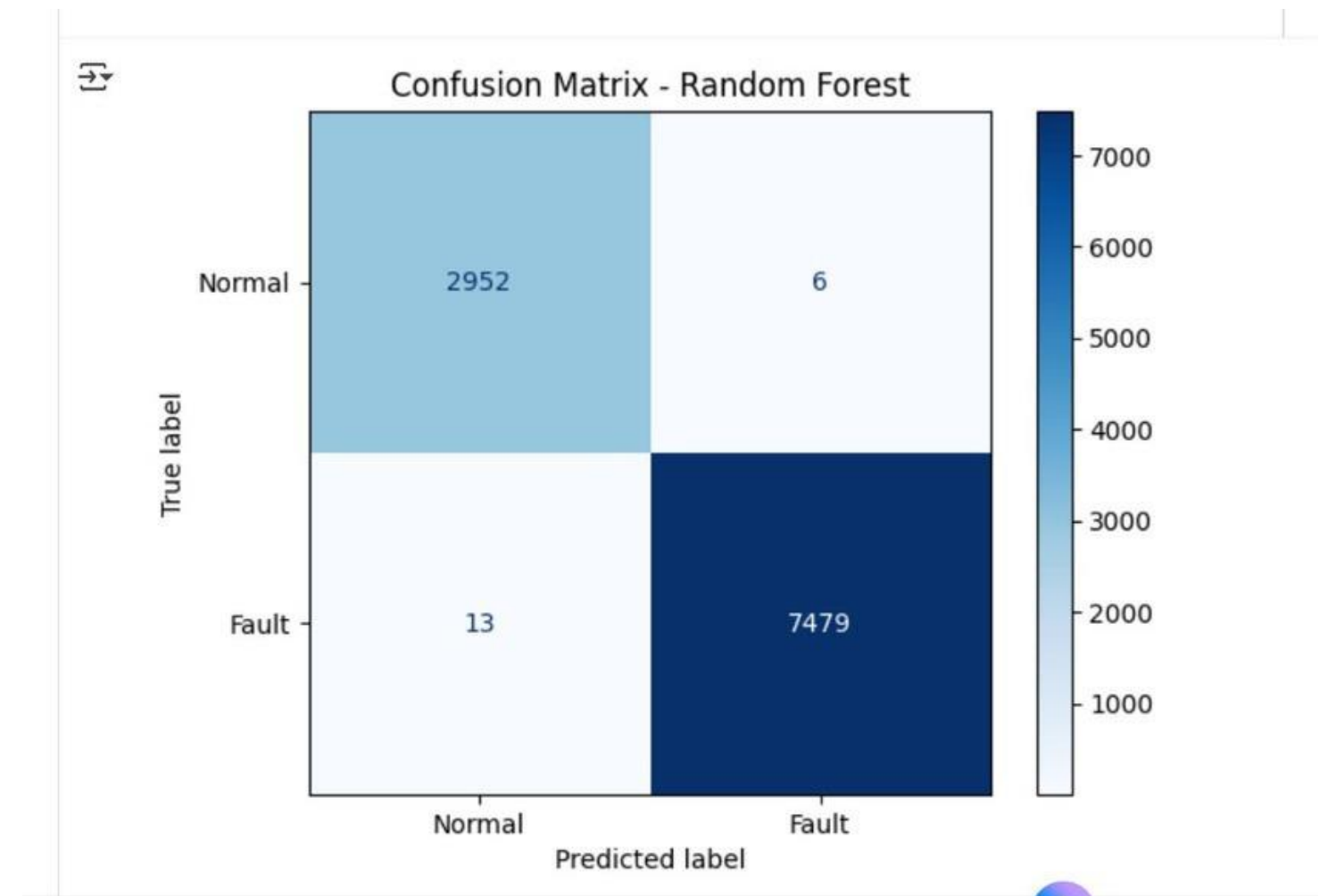
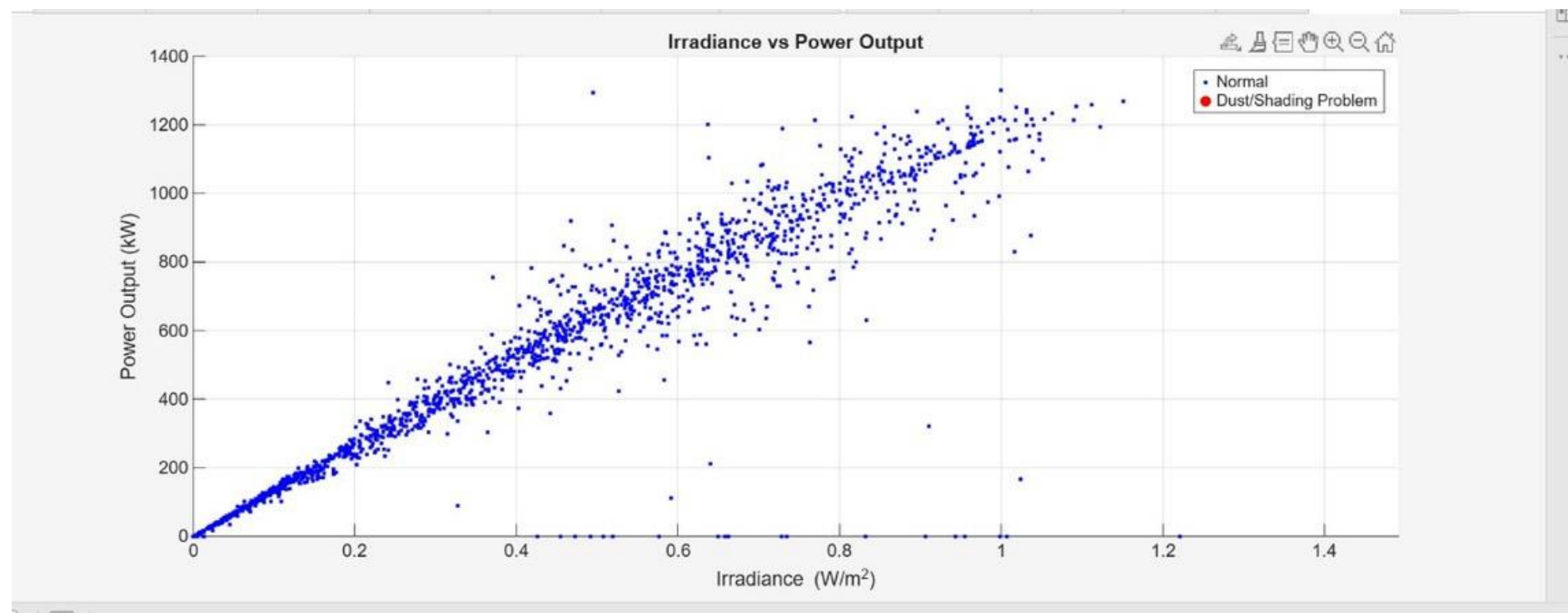
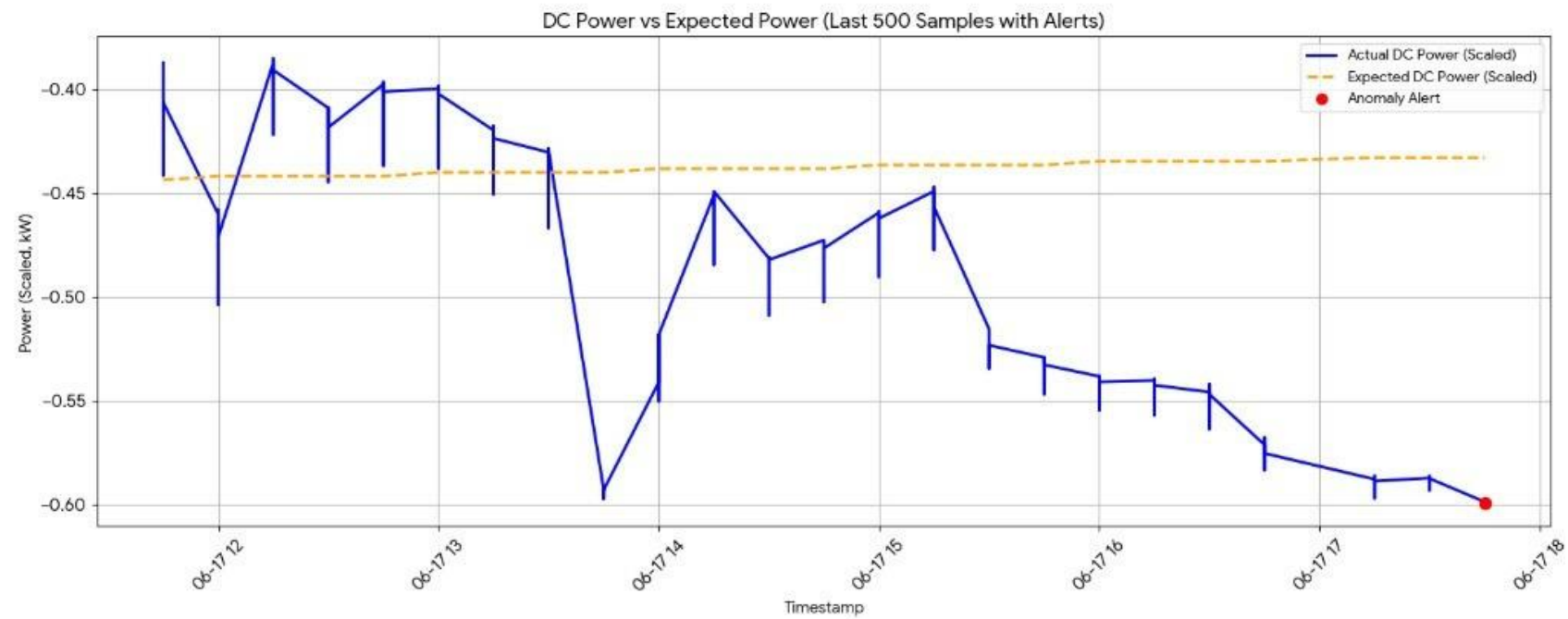
The ESP32 is powered using a USB cable or 5V supply. The ESP32 reads all sensor values and sends the data to Firebase through Wi-Fi.

The dashboard shows the live values like voltage, current, temperature, power, and alerts.



GRAPHICAL ANALYSIS

Gradient Boosting: 96.1% accuracy (best for forecasting) **Random Forest: 94.6% accuracy (stable)** **Isolation Forest: 92.1% accuracy (excellent for fault detection)**



1. Actual vs Expected Power Graph

Shows power drop or mismatch Helps detect faults or degradation Important for predictive maintenance

2. Irradiance vs Power Graph

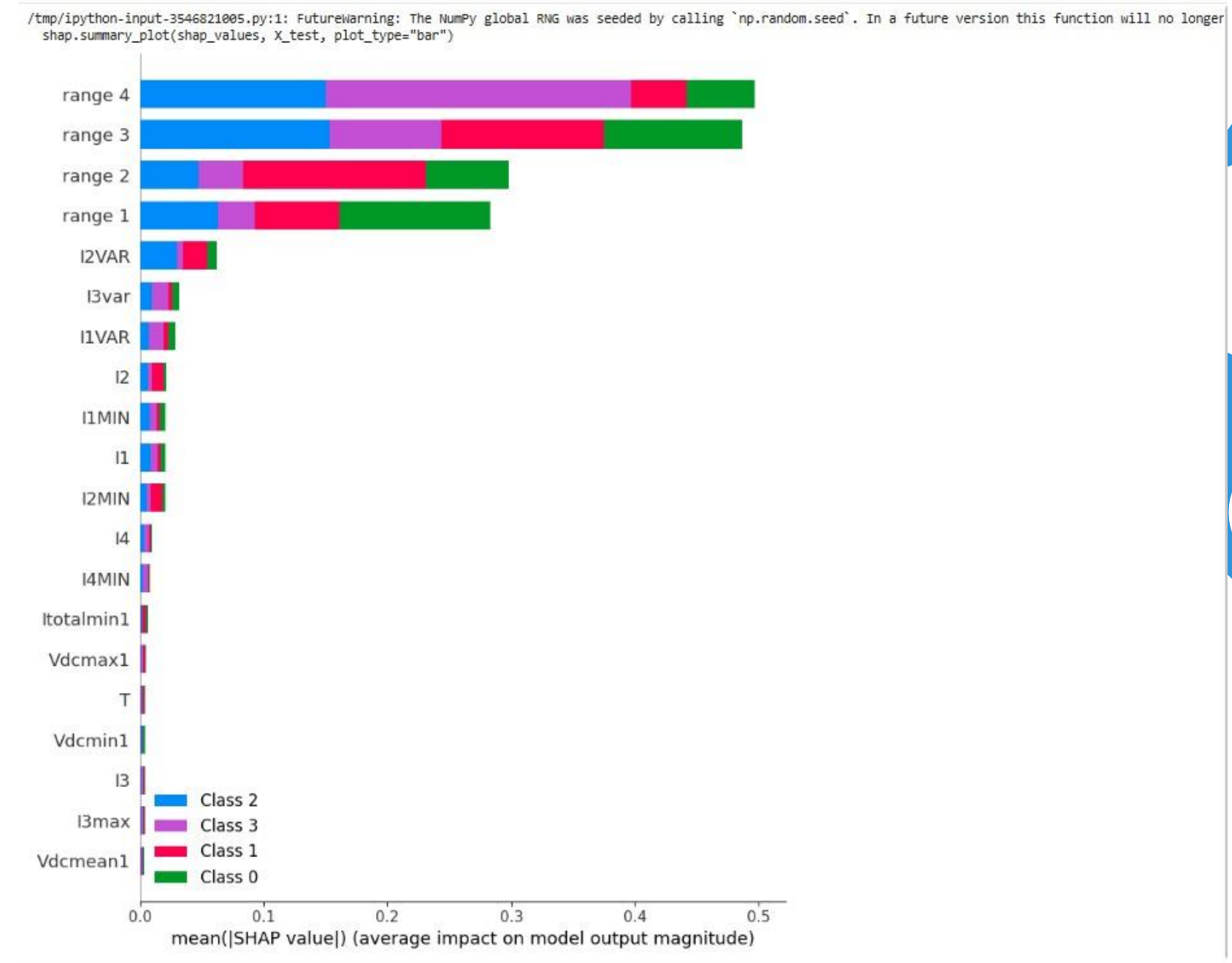
Shows how power depends on sunlight Helps understand panel performance Sensor data verification

3. Confusion Matrix

A confusion matrix is a table that shows how many predictions a model got right and wrong by comparing the actual class with the predicted class.
the model's performance better than accuracy alone.

4. SHAP Feature Importance Graph

Shows which features affect the ML model the most
Proves your model is explainable Very useful for viva and evaluation

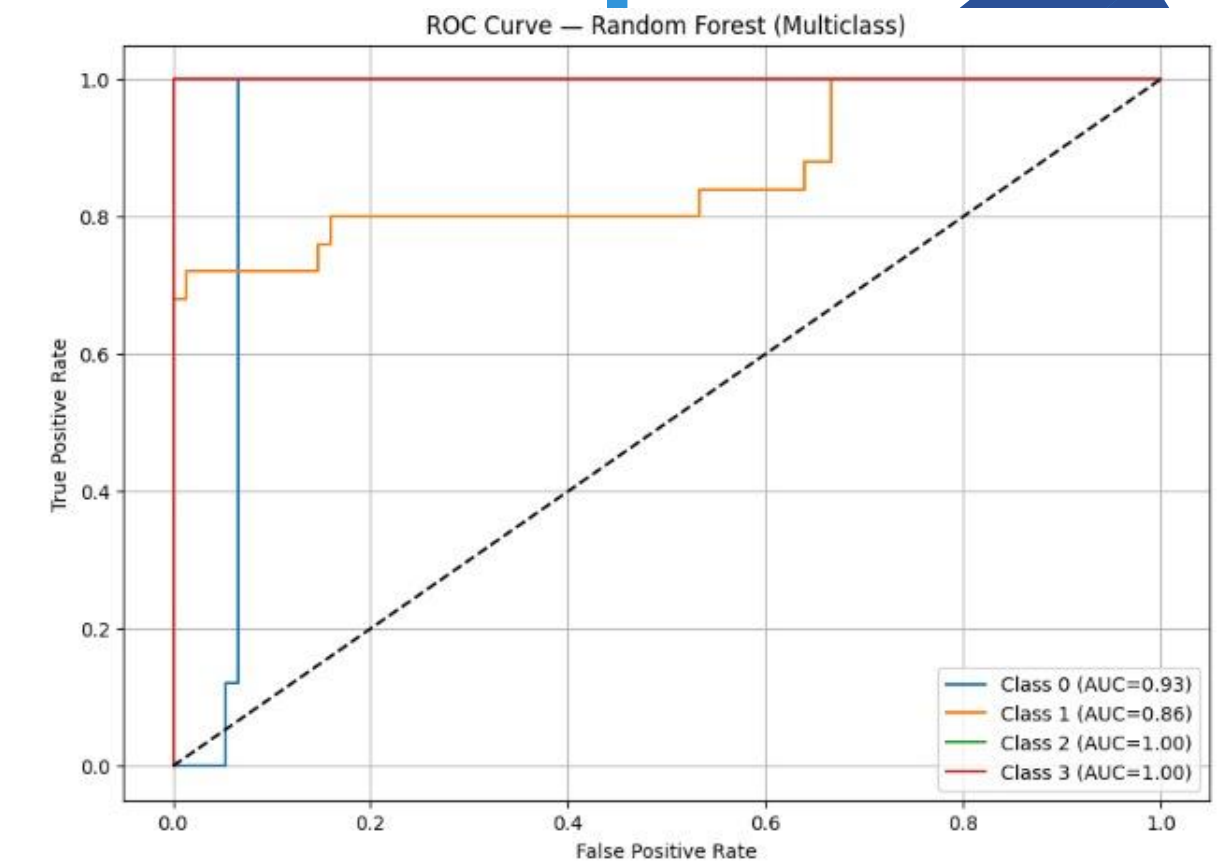
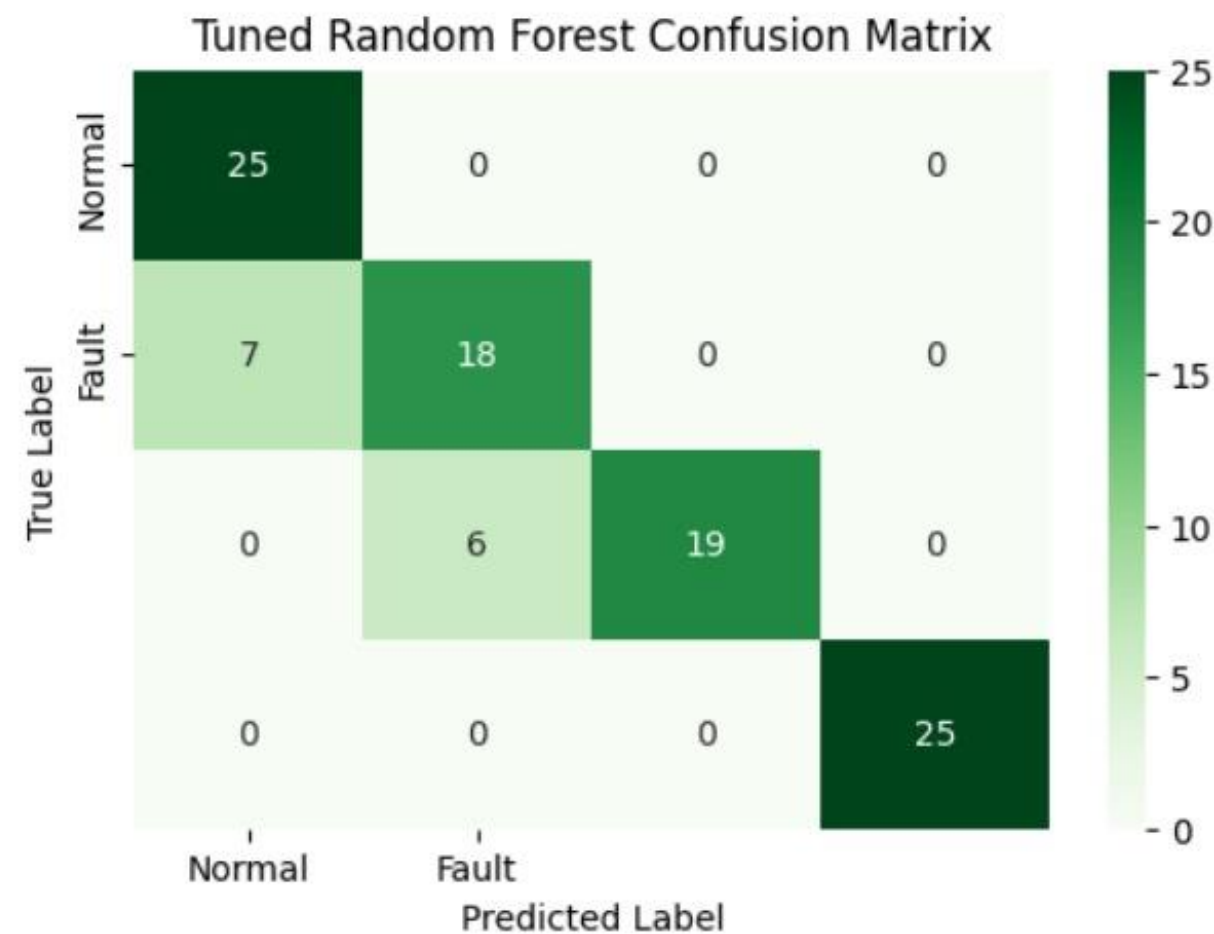


RESULT

Confusion Matrix

A confusion matrix shows how many predictions the model got correct and wrong by comparing the actual class with the predicted class.

More values on the diagonal = better accuracy
Your matrix shows very few mistakes → strong model performance



ROC Curve

The ROC curve shows how well the model can separate different classes.

The curve closer to the top-left = better detection
AUC values near 1.0 = excellent performance

Your AUCs (0.93, 0.86, 1.00, 1.00) show the model is highly accurate

Advantages

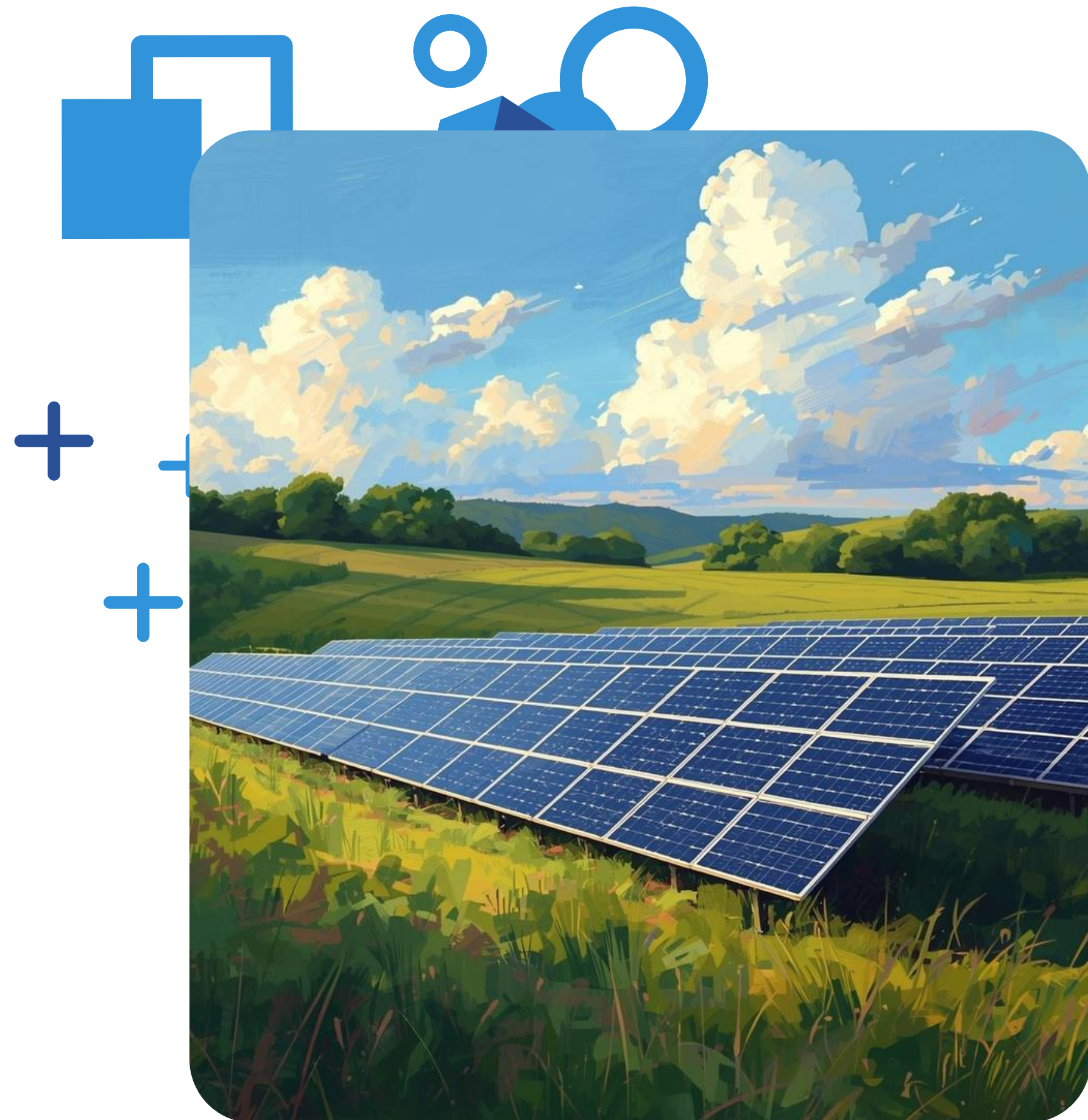
- Detects faults early and prevents system failure
- Reduces maintenance cost and manual checking
- Improves solar panel performance and power output
- Provides real-time monitoring from anywhere
- Uses machine learning for accurate fault prediction
- Cloud storage helps keep data safe and accessible
- Scalable for small and large solar systems



NOVELTY & CONTRIBUTION

Combines prediction + anomaly detection in one system Not just monitoring — system predicts issues 24–48 hours early Helps reduce maintenance cost by 25%





CONCLUSION

Successfully developed IoT+ML-based predictive maintenance system
Accurately: Predicts power output Detects anomalies in advance Helps maintain solar panels efficiently and reduces cost
Hardware+cloud give real-time monitoring

FUTURE WORK

Add camera-based fault detection
Run ML on ESP32 (TinyML)
Expand to large solar farms

Thank You!

